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Unit 4: Magnetism II

- 1 a) into the page b) to the right c) out of the page

2a)

Switch	Coil 1	Coil 2	Coil 3
closed	Counter clockwise	Counter clockwise	Counter clockwise
left closed	No change	No change	No change
Open	Clockwise	Clockwise	Clockwise

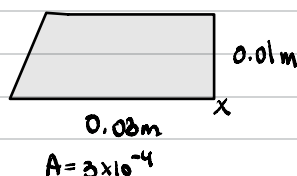
3a) 2.00 T field
ring = 2.70 cm
0.250 s to move

$$A = \pi r^2 = \frac{0.0270^2}{4} = 0.0110 \text{ m}^2$$
$$A = \pi (0.0110)^2 = 3.8 \times 10^{-4} \text{ m}^2$$
$$\text{emf}_{\text{avg}} = \frac{\Delta \Phi}{\Delta t} = \frac{7.60 \times 10^{-4}}{0.250} = 3.04 \times 10^{-3} \text{ V}$$

b) $I = \frac{\text{emf}}{R} = \frac{3.04 \times 10^{-3}}{0.0100} = 0.304 \text{ A}$

c) $P = I^2 R$
 $P = (0.304)^2 \times 0.0100 = 9.24 \times 10^{-4} \text{ W}$

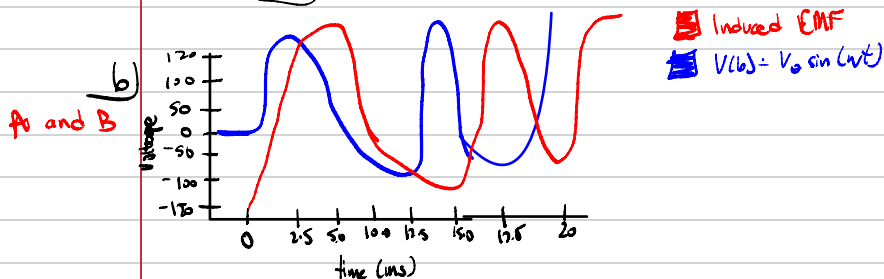
4a) $1875 \frac{\text{rad}}{\text{s}} \rightarrow 18.0 \text{ V}$
1.3 cm coil in 0.040 T field
 $V_p = N \cdot A \cdot B \cdot \omega$



$N = \frac{V_p}{A \cdot B \cdot \omega} \rightarrow N = \frac{18}{3 \times 10^{-4} (0.04)(1875)}$
 $N = 50 \text{ turns}$

b) $\omega = 2\pi f \rightarrow f = \frac{\omega}{2\pi}$
 $\frac{1875}{2\pi} = 298.4 \text{ Hz}$
 $T = \frac{1}{298.4 \text{ Hz}} = 3.35 \times 10^{-3} \text{ seconds}$

5a) 240 V to 120 V
 $\frac{240}{120} = \frac{2}{1}$
b) $\frac{120}{240} = \frac{1}{2}$



c) $V_0 = 120 \text{ V}$ $\omega = (240\pi) \text{ Hz}$ $V(t) = 0$
 $V(t) = 120 \sin(240\pi t)$
 $t = \frac{1}{120} \text{ seconds}$

7. 0.100 A, 2.00 mH, 500 V
 $\text{Emf} = L \Delta i \rightarrow \Delta t = L \frac{\Delta i}{\text{Emf}}$
 $\Delta t = \frac{2 \times 10^{-3} (0.1)}{500} = 4 \times 10^{-7} \text{ s}$

8a) $2S \cdot \Phi = L$
 $\text{EMF} = -L \frac{\Delta \Phi}{\Delta t} = -2S \times \frac{0.100}{80 \times 10^{-3}}$
 $\text{EMF} = 21250 \text{ V or } 21.25 \text{ kV}$

8b) $W = \frac{1}{2} L I^2$
 $\frac{1}{2} \cdot 2S (100)^2 = 125 \text{ kJ}$

c) $P = \frac{W}{t} = \frac{125 \times 10^{-3}}{80 \times 10^{-3}} = 1.56 \text{ MW}$

9) $L = 2S H \quad A = 0.01 \text{ m}^2 \quad l = 0.5 \text{ m} \quad N = x$
 $L = \frac{\mu_0 N^2 A}{l}$

$2S = \frac{(4\pi \times 10^{-7}) N^2 (0.01 \text{ m}^2)}{0.5 \text{ m}} = N^2 = \frac{2S \times 0.5}{(4\pi \times 10^{-7}) \cdot 0.01} = \sqrt{994719394.3} = 31539.15$

e) $X_L = 2\pi f L = 2\pi \cdot 10^4 \cdot 2S = 1.57 \times 10^6 \Omega$

9. A) $20.0 \text{ ns}, 5.0 \text{ M}\Omega$

$\tau = \frac{L}{R} = \frac{L}{R} \Rightarrow (5 \times 10^6) 20 \times 10^{-9} = 0.1 \text{ H}$

B) $R = \frac{L}{\tau} \rightarrow \frac{0.1}{1 \times 10^{-8}} = 1 \times 10^8 \Omega$

C) $I(t) = I_0 e^{-t/\tau} \rightarrow e^{-t/\tau} = \frac{I(t)}{I_0} = e^{-3/0.1} = e^{-30} = e^{-0.15} = 0.861 \rightarrow 86.1\%$

10. A) $V_{in} = iR$

B) $V_{out} = V_C$

C) $\frac{V_{out}}{V_{in}} = \frac{1}{1 + j\omega RC}$

D) $W = 2\pi f = 2\pi \cdot 100 \cdot \frac{1}{2\pi RC} = \frac{100}{RC}$

$\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + \omega^2}} = 0.999995 \rightarrow 0.01$

$\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + 0.1^2}} = 0.995$

11) A) $R = 0.1 \text{ k}\Omega$ ($C = 1 \mu\text{F}$, $L = 10 \text{ mH}$)
 1×10^{-6} 10×10^{-3}

$f_0 = \frac{1}{2\pi\sqrt{LC}}$

$F_0 = \frac{1}{2\pi\sqrt{10 \times 10^{-3} \text{ H} \times 1 \times 10^{-6} \text{ F}}} = 1,592.36 \text{ Hz}$

b) $Z = \sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2}$ $f = 0.1 f_0 \times 1592.36 = 159.236 \text{ Hz}$

$\sqrt{(0.1 \times 10^3)^2 + (2\pi \times 159.236 \times 10 \times 10^{-3} - \frac{1}{2\pi \times 159.236 \times 10^{-6}})^2} = 0.1 f_0 = 995.03 \text{ Hz}$

12 a) $V_{rms} = 120 \text{ V}$ $R = 0.1 \text{ k}\Omega$ $C = 1 \mu\text{F}$ $L = 10 \text{ mH}$
 $I_{rms} = \frac{V_{rms}}{Z} = \frac{120}{995} = 0.1206 \text{ A}$ or 120.6 mA

b) $P_{rms} = V_{rms} \cdot I_{rms} \cdot \cos \theta$

$\theta = \frac{R}{Z} = \frac{100}{995} = 0.1005 \Rightarrow P_{rms} = 120 \cdot 0.1206 \cdot 0.1005 = 1.45 \text{ W}$

13) $f_c = 1.4 \text{ MHz}$

$f_c + f_m = 1.4 \text{ MHz} + 10 \text{ kHz} = 1.410 \text{ MHz}$

$f_c - f_m = 1.4 \text{ MHz} - 10 \text{ kHz} = 1.390 \text{ MHz}$

Unit 5

1. a) B-Field km laterally, charges 0-200 nC in 2µs
 $B = \frac{\mu_0 I_0}{2\pi r}$ $I_0 = \frac{AQ}{At}$ $AQ = 200 \text{ nC} = 2 \times 10^{-7} \text{ C}$ $r = 1 \text{ cm} = 1 \times 10^{-2} \text{ m}$
 $At = 2 \mu\text{s} = 2 \times 10^{-6} \text{ s}$

$$B = (4\pi \times 10^{-7}) \left(\frac{2 \times 10^{-7}}{2 \times 10^{-6}} \right) = \frac{4 \times 10^{-7} (0.1)}{2 \times 10^{-2}} = 2.0 \times 10^{-6} \text{ T} = 2.0 \mu\text{T}$$

- b) What current is responsible for B-field?

↳ instead of conduction current its Displacement Current, still generates a magnetic field.

$$I_d = \frac{AQ}{At} = \frac{2 \times 10^{-7}}{2 \times 10^{-6}} = 0.1 \text{ A}$$

- 2a) 10 µs what displacement

$$D = c \cdot \Delta t \quad c = \text{Speed of light} = 3 \times 10^8 \frac{\text{m}}{\text{s}} \quad \Delta t = 10 \mu\text{s} = 10 \times 10^{-6} \text{ s}$$

$$D = 3 \times 10^8 (10 \times 10^{-6})$$

$$D = 3000 \text{ m round trip} \Rightarrow \frac{1}{2} \text{ one way} = 1500 \text{ m}$$

- b) What fraction of power, standard index of reflection at RF Fresnel reflection: $R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$

Standard RF reflection:

$$\begin{aligned} -\text{Air} &= 1.00 (n_1) & R &= \left(\frac{1 - 1.78}{1 + 1.78} \right)^2 = \left(\frac{-0.78}{2.78} \right)^2 = (-0.2806)^2 = 0.0787 \\ -\text{ice} &= 1.78 (n_2) \end{aligned}$$

7.87% of power is reflected

- 3a) 1 Kw onto 10 cm² at distance of 1m. Intensity?

$$I = \frac{P}{A} = \frac{1000 \text{ W}}{10 \times 10^{-4} \text{ m}^2} = 1.0 \times 10^6 \frac{\text{W}}{\text{m}^2}$$

- b) $I_2 = I_1 \left(\frac{r_1}{r_2} \right)^2$ $r_1 = 1 \text{ m}$ $r_2 = 1 \text{ cm}$ $I_1 = 1 \times 10^6 \text{ W/m}^2$
 $I_2 = 1.0 \times 10^6 \cdot \left(\frac{1}{0.01} \right)^2$ $I_2 = 2.5 \times 10^5 \text{ W/m}^2$
 $I_2 = 1 \times 10^6 \left(\frac{1}{4} \right)$

4. A) Prove that $\theta_3 = \theta_1$ in fig. 4. if $n_1 = n_3$

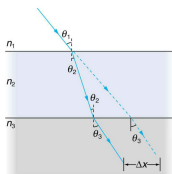


Figure 4: A light ray is in a medium with n_1 , then enters a medium with n_2 , then a medium with n_3 .

$$\theta_3 = \theta_1 \text{ if } n_1 = n_3$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_2 \sin \theta_2 = n_3 \sin \theta_3$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3$$

$$\text{if } n_1 \sin \theta_1 = n_3 \sin \theta_3, \quad n_1 = n_3. \quad \text{Then, } \sin \theta_1 = \sin \theta_3. \quad \text{So, since } n_1 = n_3, \quad \theta_1 = \theta_3$$

- b) 12 cm lens 8 cm from ear, the magnification is?

$$m = \frac{-d_i}{d_o} \quad \frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$x^2 = \left(\frac{1}{12} \right) - \left(\frac{1}{8} \right) = \frac{2}{24} - \frac{3}{24} = \frac{-1}{24} = -24 \text{ cm} = d_i$$

$$\frac{1}{f} = \frac{1}{12} \quad d = 8, \quad \frac{1}{d_o} = \frac{1}{8}$$

$$m = \frac{(-24 \text{ cm})}{8} = 3.00 \times \text{magnification}$$

- 5) 200 barns at 10 KeV, 1 cm thick, what fraction?

$$I = I_0 e^{-n\sigma x} \quad n = \text{density}, \sigma = \text{cross section}, x = \text{thickness}$$

$$\sigma = 200 \text{ barns} = 200 \times 10^{-28} \text{ m}^2 = 2.0 \times 10^{-26} \text{ m}^2$$

$$x = 0.01 \text{ m}$$

$$n = \frac{PNA}{A} = \frac{11.340 \text{ kg/m}^3 (6.022 \times 10^{23} \text{ atoms/mol})}{0.020 \text{ kg/mol}} = 3.29 \times 10^{28} \text{ atoms/m}^3 \Rightarrow \frac{I}{I_0} = e^{-3.29 \times 10^{28} \cdot 2.0 \times 10^{-26} \cdot 0.01} = e^{-6.58} = 1.39 \times 10^{-3}$$

0.139% passes through

b. A) 250 mJ into 60kg. is it radioactive?

$$1 \text{ rad} = \frac{0.01 \text{ J}}{1 \text{ kg}} = 0.01 \text{ J/kg} \quad 250 \text{ mJ} = 0.25 \text{ J}$$

$$\text{Dose} = \frac{E}{m} = \frac{0.25}{60} = 0.00417 \text{ J/kg}$$

$$\text{dose in rads} = \frac{0.00417}{0.01} = 0.417 \text{ rad}$$

b) concentration in 2kg of tissue

$$250 \text{ mJ} = 0.25 \text{ J} \quad \text{dose} = \frac{0.25}{2} = \frac{0.125}{0.01} = 12.5 \text{ rads}$$

c) RBE = 1, dose?

$$\text{Dose in Sv} = \text{Dose in Gy} \times \text{RBE}$$

$$* 1 \text{ Gy} = 100 \text{ rads}$$

$$\text{Dose in Sv} = 0.125 \times 1 = 0.125 \text{ Sv}$$

$$0.125 \text{ Gy} = 12.5 \text{ rads}$$

$$\text{RBE} = 1$$

D) No.